MEDICAPS UNIVERSITY, INDORE

Department of Computer Science & Engineering

PROJECT REPORT

GLOBAL FUEL PRICES ANALYSIS

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TABLE OF CONTENTS

1. INTRODUCTION

1.1 Project Overview

1.2 Purpose

2. IDEATION PHASE

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstorming

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

3.2 Solution Requirement

3.3 Data Flow Diagram

3.4 Technology Stack

4. PROJECT DESIGN

4.1 Problem Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

7. RESULTS

7.1 Output Screenshots

8. ADVANTAGES & DISADVANTAGES

9. CONCLUSION

10. FUTURE SCOPE

11. APPENDIX

Source Code (if any)

Dataset Link

GitHub & Project Demo Link

1. INTRODUCTION

1.1 Project Overview

Global Fuel Prices Analysis is a data-visualization project that explores petrol-price differences across countries and examines how fuel prices relate to factors such as fuel taxes and Brent crude oil prices. The project uses a fuel-price dataset and Tableau to turn tabular data into charts, dashboards, and a published Story. Three dashboards—Policy, Logistics, and Research—organize the analysis for different types of users.

The dashboards are intended to make comparisons easier than reading rows of raw data. Users can explore patterns across countries and categories, subject to the fields and filters available in the final workbook.

1.2 Purpose

- To present fuel-price data in a clear and visual format.
- To compare petrol prices across countries and identify broad differences.
- To explore tax bands and price categories using calculated fields.
- To examine the relationship between petrol prices and Brent crude oil prices where the dataset supports the comparison.
- To publish the dashboards as a Tableau Story and maintain supporting project files in GitHub.

2. IDEATION PHASE

2.1 Problem Statement

Fuel prices differ across countries, and the raw dataset may be difficult for a reader to interpret quickly. Comparing countries, tax levels, and crude-oil-related measures manually can take time and make patterns harder to notice. The project addresses this problem by organizing the available data into Tableau visualizations and dashboards so users can explore the information more easily.

2.2 Empathy Map Canvas

The following empathy map describes likely users of a fuel-price analysis dashboard. It is a design aid for the project, not the result of a formal interview.

Empathy area	Likely user perspective
Says	“I want to compare fuel prices between countries without reading a large spreadsheet.”
Thinks	“Which countries have higher prices, and how might tax levels relate to them?”
Does	Uses charts, compares countries or categories, and checks dashboard filters.
Feels	May feel confused by raw rows and prefers clear labels, summaries, and visual comparisons.
Needs / pains	Needs understandable charts, consistent units, clear categories, and reliable data; raw data can be time-consuming to compare.

2.3 Brainstorming

During ideation, possible ways to present the data were considered. The selected approach was to group visualizations into dashboards for policy, logistics, and research perspectives.

- Compare petrol prices by country or region.
- Group observations into low, medium, and high tax bands.
- Classify petrol prices into price categories for easier scanning.

- Explore petrol price in relation to Brent crude oil price.
- Publish a Tableau Story so viewers can navigate the project online.
- Keep the dataset, workbook, documentation, and related files organized in GitHub.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

This journey map represents a typical viewer exploring the published Tableau Story.

Stage	User action	Possible difficulty	Project support
Discover	Opens the shared Tableau Story link	Unsure what the project covers	Project title and overview explain the topic
Explore	Opens Policy, Logistics, or Research dashboard	Many values can be hard to compare	Charts group related information
Filter / compare	Selects available filters or compares marks	May need to interpret categories	Labels, legends, and available filters guide exploration
Understand	Reviews patterns in fuel prices and tax categories	Correlation may be mistaken for causation	Report explains that charts show relationships, not proof of cause
Share	Uses the published Story or repository files	May need source files or methodology	Tableau link and GitHub repository provide access, once links are added

3.2 Solution Requirement

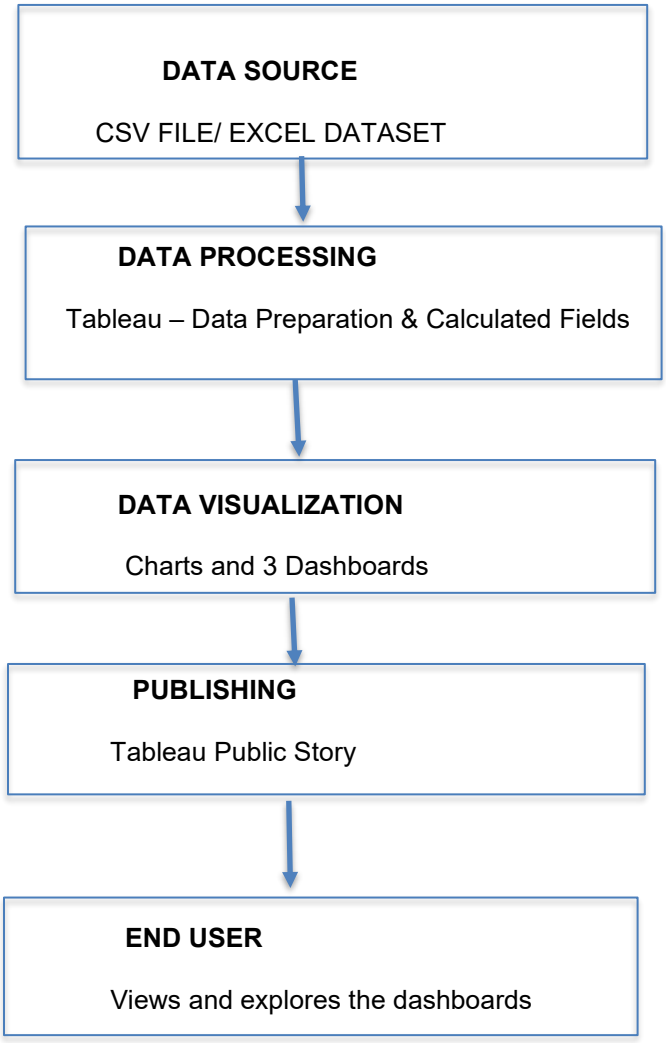
Functional requirements

- The project should load the prepared CSV or Excel data into Tableau.
- It should provide visual comparisons of petrol prices across countries.
- It should include the Policy, Logistics, and Research dashboards.
- It should use calculated fields for tax bands, price categories, and the petrol-to-Brent ratio where the corresponding fields exist.
- It should publish a Story that can be opened by a viewer.
- It should provide supporting documentation and project files through GitHub.

Non-functional requirements

- Charts should have readable titles, labels, and legends.
- Visualizations should be reasonably responsive for normal browsing.
- Data fields and units should be used consistently.
- Links should be accessible to the intended audience and the published Story should open correctly.
- Findings should be based on the displayed data; no unverified numerical claims should be included.

3.3 Data Flow Diagram



Explanation: The dataset is the input. Tableau is used to prepare fields and create calculated measures. The resulting charts are arranged into dashboards and published as a Story for viewers. GitHub stores supporting project files and documentation; it is not the visualization engine.

3.4 Technology Stack

Technology / file type	Purpose
Tableau Desktop / Tableau Public	Create charts, dashboards, calculated fields, and publish the Story
CSV dataset	Tabular input data for analysis
Excel (.xlsx)	Spreadsheet version of the project dataset
GitHub	Versioned hosting for supporting files and README

Tableau Public	Online access to the published Story
DOCX / PDF report	Project documentation and final submission record

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem is that raw fuel-price data can be difficult to compare and interpret. Tableau charts and dashboards provide a visual interface that helps users compare countries and examine tax and crude-oil-related measures. The solution fits the problem because it changes a large table into organized views. Its usefulness depends on data quality, appropriate chart selection, readable labels, and correct interpretation.

4.2 Proposed Solution

The proposed solution is an interactive Tableau Story named Global Fuel Prices Analysis. It contains three dashboards: Policy, Logistics, and Research. The project uses calculated fields such as Tax Band and Price Category to group observations, and Price Brent Ratio to compare average petrol price with average Brent crude price. The final published workbook should be treated as the source of truth for exact chart titles, filters, and formulas.

4.3 Solution Architecture

1. Data input: prepared CSV and Excel files.

2. Data connection: load the dataset into Tableau and verify field types.

3. Analysis: create calculated fields and build charts.

4. Presentation: arrange charts into Policy, Logistics, and Research dashboards.

5. Publishing: publish the Story to Tableau Public.

6. Documentation: keep the workbook, dataset where permitted, README, and report in GitHub.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Main tasks	Output
1. Define the problem	Identify the need for comparing fuel prices	Problem statement
2. Prepare data	Review the CSV/Excel files and check fields	Dataset ready for Tableau
3. Analyze data	Create calculated fields and explore the measures	Calculated fields and analysis
4. Build visualizations	Create charts and organize three dashboards	Policy, Logistics, Research dashboards
5. Publish	Publish the Story and check the public link	Tableau Public Story
6. Document and submit	Prepare report, organize files, and add repository/demo links	Final submission package

Note: This is a phase-based plan. Add actual dates or duration only if required by the project supervisor and based on the work completed.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The following checklist can be used to validate the final workbook. Mark each test as Pass or Fail after checking the published Story and local workbook.

Test	How to check	Status
Data connection	Open the workbook and confirm the data source loads without an error	Pass
Calculated fields	Check formulas and category thresholds against the workbook	Pass
Dashboard display	Open all three dashboards and confirm charts render	Pass
Filters / interactions	Try each configured filter and confirm charts respond correctly	Pass
Labels and units	Check titles, legends, labels, and units for readability	Pass
Published link	Open the Tableau Public Story link in a browser	Pass
Responsiveness	Navigate between dashboards and observe whether they load smoothly	Pass

No measured load-time or benchmark result is claimed in this report. Record actual observations if formal performance measurements are required.

7. RESULTS

The project produces a published Tableau Story with three dashboards—Policy, Logistics, and Research—and visualizations intended to support comparisons of fuel prices, tax categories, and related measures. The exact conclusions should be taken from the final published workbook after checking the charts.

7.1 Output Screenshots

Insert screenshots from the final Tableau workbook in the spaces below. Use screenshots of the actual published dashboards; do not use placeholder images as final evidence.

Figure 1: Policy Dashboard

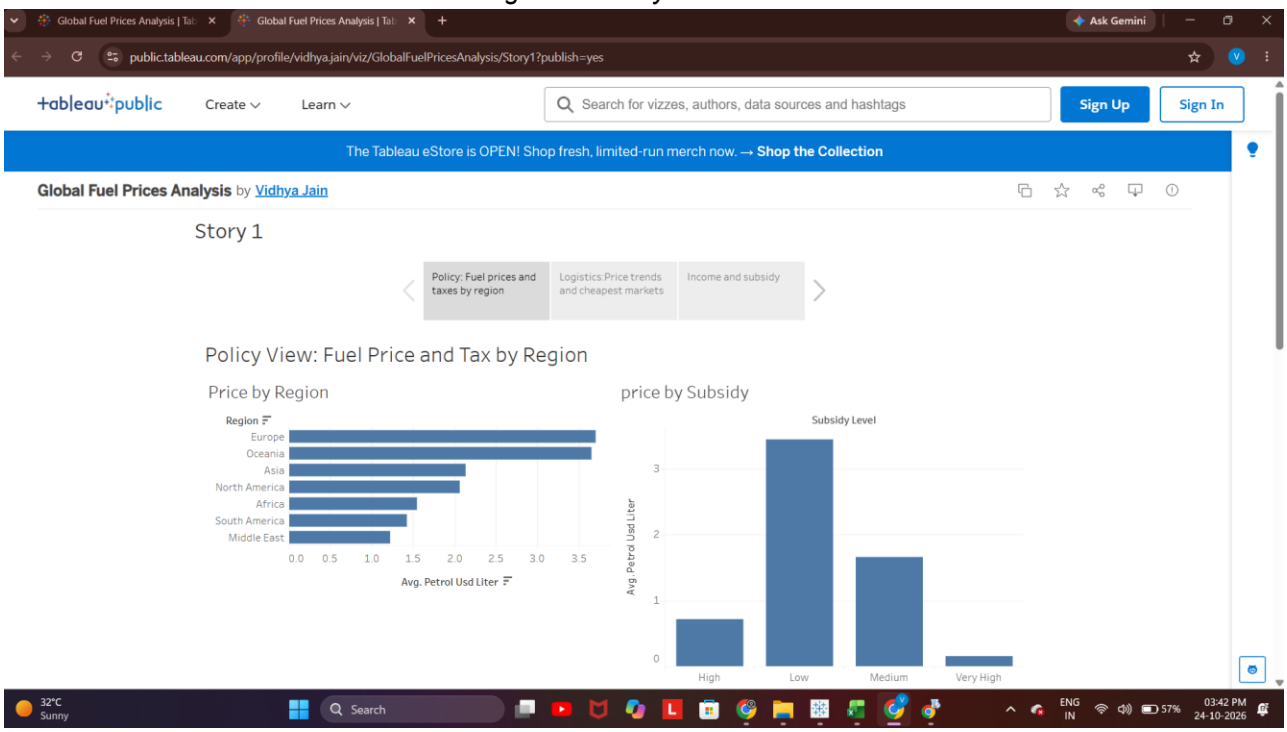


Figure 2: Logistics Dashboard

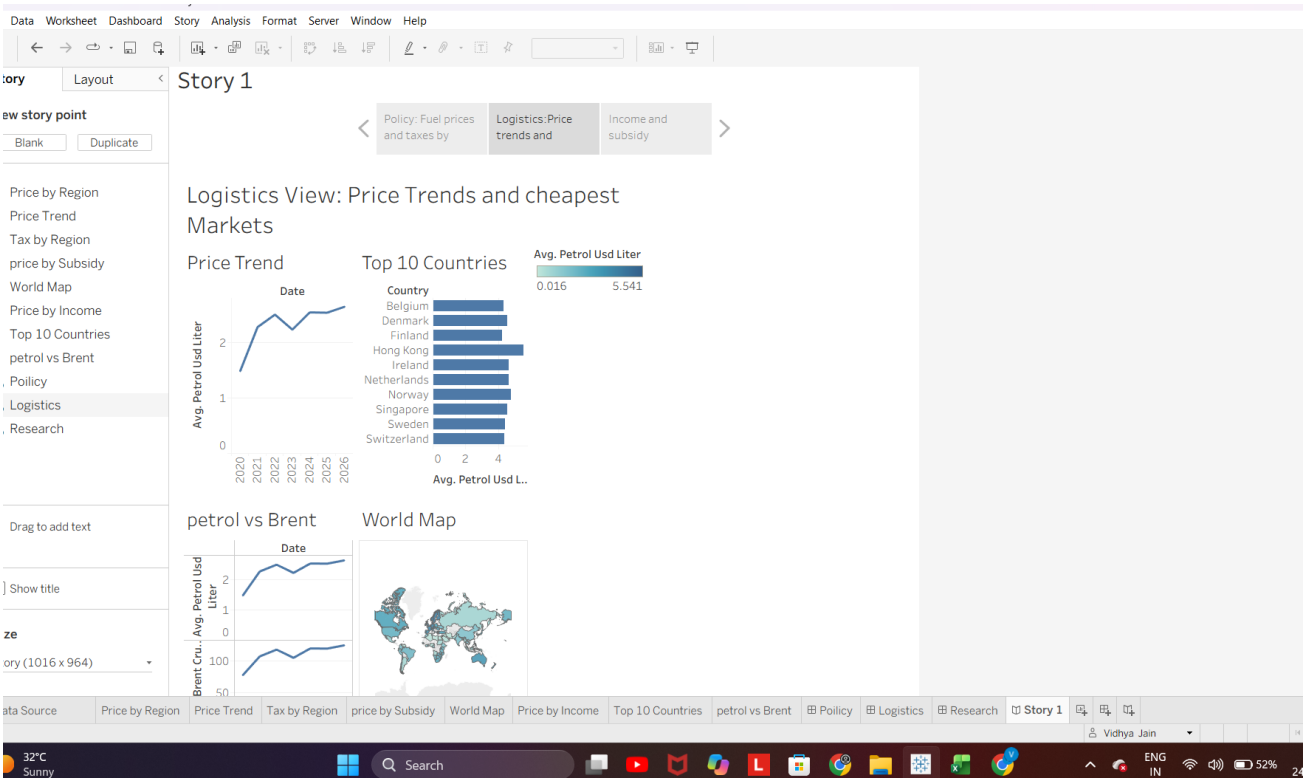
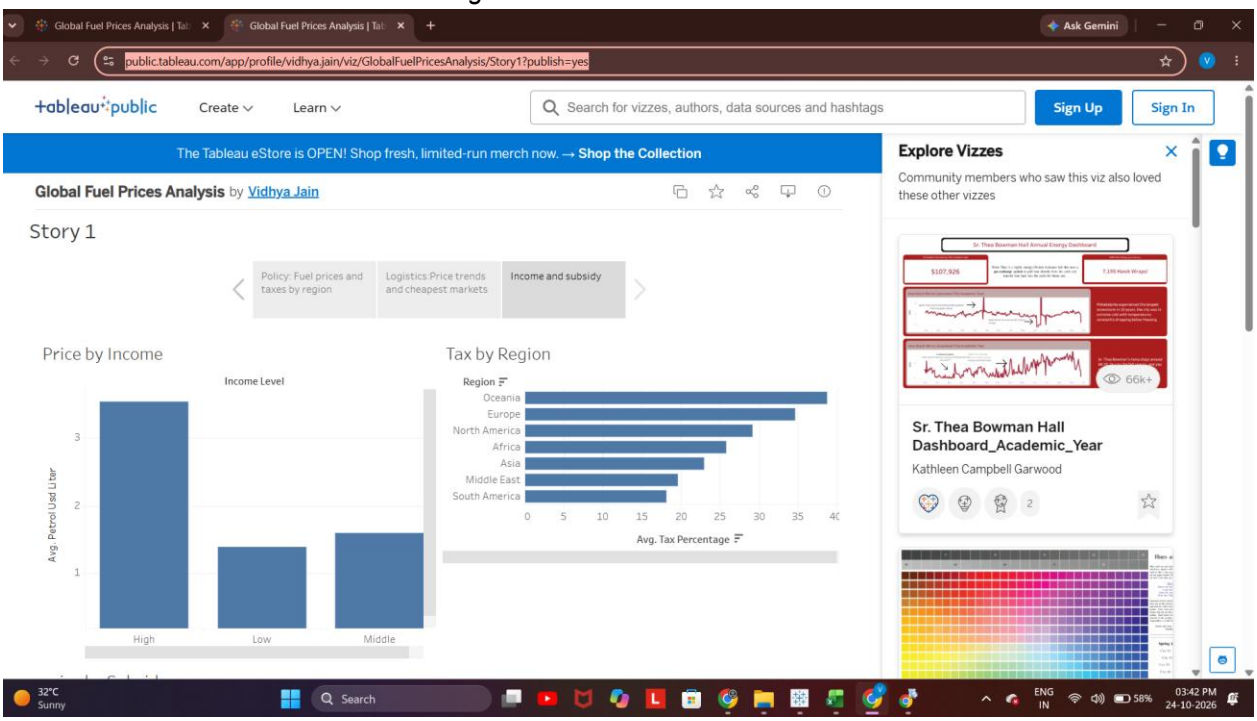


Figure 3: Research Dashboard



8. ADVANTAGES & DISADVANTAGES

Advantages

- Visual comparison is easier than reviewing raw spreadsheet rows.
- Dashboards organize the analysis into different perspectives.
- Calculated categories can help viewers scan patterns quickly.
- The published Story can be shared through a web link.
- Supporting files can be organized in a GitHub repository.

Disadvantages / Limitations

- The analysis is limited by the completeness, accuracy, and coverage of the available dataset.
- Charts show patterns and associations; they do not by themselves prove that one factor causes another.
- Interpretation depends on correct units, field definitions, and calculated-field thresholds.
- Viewers may need an internet connection to open the published Tableau Story.
- Specific numerical findings must be verified against the final workbook before being reported.

9. CONCLUSION

Global Fuel Prices Analysis demonstrates how Tableau can be used to transform a fuel-price dataset into visual dashboards and a shareable Story. The Policy, Logistics, and Research dashboards provide structured views for exploring country-level price differences and related measures. The project also demonstrates the use of calculated fields and online publication. Before final submission, the dataset source, actual dashboard screenshots, verified findings, GitHub repository URL, and any required demonstration video should be added and checked.

10. FUTURE SCOPE

- Add more recent and reliable fuel-price observations when available.
- Document the original dataset source, data definitions, units, and update frequency.
- Add more countries or historical comparisons if suitable data is available.
- Improve dashboard tooltips, filter controls, accessibility, and mobile readability.
- Include verified key findings and clearly explain limitations.
- Automate data refresh if the source and Tableau setup support it.
- Add a short demonstration video explaining the dashboards and their use.

11. APPENDIX

A. Source Code / Calculated Fields

This is a Tableau visualization project rather than a conventional software application, so there may be no separate source-code file. The calculated fields used during development are listed below; verify their exact names and formulas in the final workbook before submission.

Tax Band — intended grouping: High for tax percentage ≥ 35; Medium for tax percentage ≥ 20 and < 35; Low for tax percentage < 20.

Formula: IF [Tax Percentage] >= 35 THEN "High"

ELSEIF [Tax Percentage] >= 20 THEN "Medium"

ELSE "Low"

END

Price Category — intended grouping: Expensive for Petrol Usd Liter ≥ 3; Medium for Petrol Usd Liter ≥ 1 and < 3; Cheap for Petrol Usd Liter < 1.

Formula : IF [Petrol Usd Liter] >= 3 THEN "Expensive"

ELSEIF [Petrol Usd Liter] >= 1 THEN "Medium"

ELSE "Cheap"

END

Price Brent Ratio — AVG([Petrol Usd Liter]) / AVG([Brent Crude Usd]) * 100.

B. Dataset Link

Dataset file names used during development: fuel_prices_clean.csv and global_fuel_prices_Dataset.xlsx.

Original dataset source URL:
[https://drive.google.com/file/d/13k0qQW8OqvyIrE3yCzAym5SUTmQNd0qu/view?pli=1].

C. GitHub & Project Demo Link

Tableau Public Story:
https://public.tableau.com/app/profile/vidhya.jain/viz/GlobalFuelPricesAnalysis/Story1?publish=yes

GitHub repository: [https://github.com/vidhya-jain/fuel-prices-project].

Demo / explanation video:
[https://drive.google.com/file/d/1OFsIwGC01hxrHGgM_6wzWetBCvSMjlsk/view?usp=sharing].

D. Final Submission Checklist

- ✓ Verify all three dashboards and insert actual screenshots.
- ✓ Add the original dataset source URL.
- ✓ Verify calculated-field names, formulas, and thresholds in Tableau.
- ✓ Add two or three findings that are directly visible in the final charts.
- ✓ Paste the exact public GitHub repository URL.
- ✓ Add the demo video link if the assignment requires a video.
- ✓ Open the Tableau Story link and confirm it is accessible.